PRODUCT USER MANUAL

OMI_EXTREME_WAVE
Significant wave height

OMI_EXTREME_WAVE_BALTIC_swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ NORTHWESTSHELF_swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ IBI _swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ MEDSEA _swh_mean_and_anomaly_obs

Issue 2.1

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Approval Date: November 2025

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Change Record

Issue	Date	§	Description of Change	Author	Checked by	Validated by
	20/07/2019	all	Creation of the document	Marta de Alfonso, Alex Gallardo, Fernando Manzano		
	26/07/2019	all	Modified	Begoña Pérez Gómez		
1.0	02/10/2019	all	Adapt the document to the new OMI structure	Marta de Alfonso		
2.0	02/09/2022		New Product naming and new template	Marta de Alfonso		Stéphane Tarot
2.1	07/05/2025		Added two new variables for uncertainties	Marta de Alfonso, Begoña Pérez Gómez, Alex Gallardo		Stéphane Tarot

GLOSSARY AND ABBREVIATIONS

MDS	Marine Data Store
MOI	Mercator Océan International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Service wave extreme variability Ocean Monitoring Indicators (OMIs) :

OMI_EXTREME_WAVE_BALTIC_swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ NORTHWESTSHELF_swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ IBI _swh_mean_and_anomaly_obs
OMI_EXTREME_WAVE_ MEDSEA _swh_mean_and_anomaly_obs

These indicators provide information about the 1st and 99th percentiles of the essential ocean variable significant wave height (swh). The percentiles have been computed from in situ observations measured by buoys, mainly along the coast.

These OMIs are organized in one dataset each

- omi_extreme_wave_[region]_swh_mean_and_anomaly_obs: describes the spatial distribution of the annual percentiles (mean over a period, target year and anomaly) for the different stations.

The products are published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The products are available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF4 format and follow CF-1.6 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

More information about EXTREME WAVE OMIs can be found in the Quality Information Document (QUID) (https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-EXTREME-WAVE-OBS.pdf). See also News flash. More detailed information can be obtained from the Copernicus Marine Service Service Desk (servicedesk.cmems@mercator-ocean.eu).

Information on operational issues on products and services can be found on our **User Notification Service**. If you have any questions, please **contact us**.

I.2. History of changes

2019/10	Publication of OMIs: BALTIC_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly_obs NORTHWESTSHELF_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly_obs IBI_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly_obs MEDSEA_OMI_SEASTATE_extreme_var_swh_mean_and_anomaly_obs
2022/11	New OMIs naming: OMI_EXTREME_WAVE_BALTIC_swh_mean_and_anomaly_obs OMI_EXTREME_WAVE_ NORTHWESTSHELF_swh_mean_and_anomaly_obs OMI_EXTREME_WAVE_ IBI _swh_mean_and_anomaly_obs OMI_EXTREME_WAVE_ MEDSEA _swh_mean_and_anomaly_obs
2025/05	Estimation of uncertainties with standard deviation of percentiles. Added two new variables: swh_percentile1_std and swh_percentile99_std.

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	**OMI_EXTREME_WAVE_BALTIC_swh_mean_and_anomaly_obs**
Geographical coverage	Baltic Sea
Variables	swh_percentile1_mean swh_percentile1_targetyear swh_percentile1_anomaly swh_percentile1_std swh_percentile99_mean swh_percentile99_targetyear swh_percentile99_anomaly swh_percentile99_std
Update frequency	Yearly
Available time series	N/A (no time series but period covered: 01/01/2000 - 31/12/2023)
Delivery mechanism	MDS
Temporal/Spatial resolution	N/A
Integration depth (for example..)	Surface
Format	NetCDF4

Product(s) name	**OMI_EXTREME_WAVE_ NORTHWESTSHELF_swh_mean_and_anomaly_obs**
Geographical coverage	North West Shelf area
Variables	swh_percentile1_mean swh_percentile1_targetyear swh_percentile1_anomaly swh_percentile1_std swh_percentile99_mean swh_percentile99_targetyear swh_percentile99_anomaly swh_percentile99_std
Update frequency	Yearly
Available time series	N/A (no time series but period covered: 01/01/2000 - 31/12/2023)
Delivery mechanism	MDS
Temporal/Spatial resolution	N/A
Integration depth (for example..)	Surface

Format	NetCDF4

Product(s) name	**OMI_EXTREME_WAVE_ IBI _swh_mean_and_anomaly_obs**
Geographical coverage	Iberia Biscay Ireland Seas
Variables	swh_percentile1_mean swh_percentile1_targetyear swh_percentile1_anomaly swh_percentile1_std swh_percentile99_mean swh_percentile99_targetyear swh_percentile99_anomaly swh_percentile99_std
Update frequency	Yearly
Available time series	N/A (no time series but period covered: 01/01/2000 - 31/12/2023)
Delivery mechanism	MDS
Temporal/Spatial resolution	N/A
Integration depth (for example..)	Surface
Format	NetCDF4

Product(s) name	**OMI_EXTREME_WAVE_ MEDSEA _swh_mean_and_anomaly_obs**
Geographical coverage	Mediterranean Sea
Variables	swh_percentile1_mean swh_percentile1_targetyear swh_percentile1_anomaly swh_percentile1_std swh_percentile99_mean swh_percentile99_targetyear swh_percentile99_anomaly swh_percentile99_std
Update frequency	Yearly
Available time series	N/A (no time series but period covered: 01/01/2000 - 31/12/2023)
Delivery mechanism	MDS
Temporal/Spatial resolution	N/A
Integration depth (for example..)	Surface
Format	NetCDF4

II.2. Details of the datasets

OMI_EXTREME_WAVE_BALTIC_swh_mean_and_anomaly_obs **omi_extreme_wave_baltic_swh_mean_and_anomaly_obs** Describes the spatial distribution of the annual percentiles (mean over the historical period, target year and anomaly of the target year with respect to the mean) for the different in-situ stations along the Baltic Sea. The dataset contains one file with the following variables for each location
swh_percentile1_mean [m] Long name: “averaged 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_targetyear** [m] Long name: “1st percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_anomaly [m]** Long name: “anomaly in the 1st percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_std** Long name: “standard deviation of 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_mean** [m] Long name: “averaged 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_targetyear** [m] Long name: “99th percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_anomaly [m]** Long name: “anomaly in the 99th percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_std** Long name: “standard deviation of 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height"

OMI_EXTREME_WAVE_ NORTHWESTSHELF_swh_mean_and_anomaly_obs **omi_extreme_wave_northwestshelf_swh_mean_and_anomaly_obs** Describes the spatial distribution of the annual percentiles (mean over the historical period, target year and anomaly of the target year with respect to the mean) for the different in-situ stations along the North West Shelf area. The dataset contains one file with the following variables for each location
swh_percentile1_mean [m] Long name: “averaged 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_targetyear** [m] Long name: “1st percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_anomaly [m]** Long name: “anomaly in the 1st percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_std** Long name: “standard deviation of 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_mean** [m] Long name: “averaged 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_targetyear** [m] Long name: “99th percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_anomaly [m]** Long name: “anomaly in the 99th percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_std** Long name: “standard deviation of 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height"

OMI_EXTREME_WAVE_ IBI _swh_mean_and_anomaly_obs **omi_extreme_wave_ibi_swh_mean_and_anomaly_obs** Describes the spatial distribution of the annual percentiles (mean over the historical period, target year and anomaly of the target year with respect to the mean) for the different in-situ stations along the Iberia Biscay Ireland Seas. The dataset contains one file with the following variables for each location
swh_percentile1_mean [m] Long name: “averaged 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_targetyear** [m] Long name: “1st percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_anomaly [m]** Long name: “anomaly in the 1st percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_std** Long name: “standard deviation of 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_mean** [m] Long name: “averaged 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_targetyear** [m] Long name: “99th percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_anomaly [m]** Long name: “anomaly in the 99th percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_std** Long name: “standard deviation of 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height"

OMI_EXTREME_WAVE_ MEDSEA _swh_mean_and_anomaly_obs **omi_extreme_wave_medsea_swh_mean_and_anomaly_obs** Describes the spatial distribution of the annual percentiles (mean over the historical period, target year and anomaly of the target year with respect to the mean) for the different in-situ stations along the Mediterranean Sea. The dataset contains one file with the following variables for each location
swh_percentile1_mean [m] Long name: “averaged 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_targetyear** [m] Long name: “1st percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_anomaly [m]** Long name: “anomaly in the 1st percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile1_std** Long name: “standard deviation of 1st percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_mean** [m] Long name: “averaged 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_targetyear** [m] Long name: “99th percentile of significant wave height in the target year” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_anomaly [m]** Long name: “anomaly in the 99th percentile of significant wave height (target year – mean)” Standard name: “sea_surface_wave_significant_height" **swh_percentile99_std** Long name: “standard deviation of 99th percentile of significant wave height” Standard name: “sea_surface_wave_significant_height"

II.3. Product System Description

The OMI_EXTREME_WAVE indicators are based on the computation of the 99th and the 1st percentiles (that represent approximately the 1% highest and lowest values, respectively) from in situ data (significant wave height observations), following the same approach applied by Woodworth and Blackman (2004) to extreme sea levels from tide gauges. This method determines changes in the frequency distribution of the measured variables.

All the data used for the computation of the extremes variability indicators have been measured by buoys and have been downloaded from Copernicus Marine Service In Situ Thematic Assembly Centre (In Situ TAC) products. The analysis is performed for the following variable: significant wave height (swh).

The periods selected are based on our analysis of the data available in the different In Situ TAC regions that fulfilled specific criteria of data completeness and quality, as described in Pérez Gómez et al 2016 and Pérez Gómez et al 2018. Buoys were selected according to consolidated time series of hourly data in Copernicus Marine Service for the period: 2000 to the target year. Within this period, a minimum number of 10 years with completeness index larger than 70% is required for a station to be included. These numbers can be reduced to 8 years and 60% in case that the coverage is not enough to make the OMIs properly representative.

These data are used to compute annually the 99th and 1st percentile values at each station. Then, the analysis to display the spatial variability of the mean and anomaly (products: OMI_EXTREME_WAVE_[region]_swh_mean_and_anomaly_obs) is performed based on the computation for each station of the anomaly of both percentiles, for the target year, with respect to their historical means. This analysis allows to check the geographical coherence of the OMI and identify local variability in the extremes. The uncertainty for each percentile is given by the standard deviation included in the file. The areas or regions displayed in these OMIs correspond mainly to the same geographical areas of the Copernicus Marine Service In Situ TAC.

III FILES NOMENCLATURE AND FORMAT

III.1. Nomenclature of files

The Copernicus Marine Service sea state OMI files nomenclature when downloaded through the Copernicus Marine Service MDS Services is based on the following nomenclature:

Datasets omi_extreme_wave_[region]_swh_mean_and_anomaly_obs :

<datasetname>_<freetext>.nc

Where :

- **<datasetname>** is the dataset reference**.**
- **<freetext>** can include potential additional information.

Example: omi_extreme_wave_baltic_swh_mean_and_anomaly_obs.nc

III.2. File Format

The products are stored using the NetCDF format.

NetCDF (network Common Data Form) is an interface for array-oriented data access and a library that provides an implementation of the interface. The NetCDF library also defines a machine-independent format for representing scientific data. Together, the interface, library, and format support the creation, access, and sharing of scientific data. The NetCDF software was developed at the Unidata Program Center in Boulder, Colorado. The NetCDF libraries define a machine-independent format for representing scientific data.

Please see Unidata NetCDF pages for more information, and to retrieve NetCDF software package.

NetCDF data is:

- Self-Describing. A netCDF file includes information about the data it contains.
- Architecture-independent. A NetCDF file is represented in a form that can be accessed by computers with different ways of storing integers, characters, and floating-point numbers.
- Direct-access. A small subset of a large dataset may be accessed efficiently, without first reading through all the preceding data.
- Appendable. Data can be appended to a NetCDF dataset along one dimension without copying the dataset or redefining its structure. The structure of a NetCDF dataset can be changed, though this sometimes causes the dataset to be copied.

Sharable. One writer and multiple readers may simultaneously access the same NetCDF file.

III.3. Reading Softwares

The products are stored using the NetCDF-CF format. Note that these data can be browsed and used through several softwares, like:

- ✓ Basic Radar Altimetry Toolbox: http://www.altimetry.info
- ✓ ncBrowse: http://www.epic.noaa.gov/java/ncBrowse/
- ✓ NetCDF Operator (NCO): http://nco.sourceforge.net/
- ✓ http://www.unidata.ucar.edu/software/netcdf/

IVREFERENCES

Quality Information Document (QUID) CMEMS-OMI-QUID-EXTREME-WAVE-OBS (https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OMI-QUID-EXTREME-WAVE-OBS.pdf).

Pérez-Gómez B, Álvarez-Fanjul E, She J, Pérez-González I, Manzano F. 2016. Extreme sea level events, Section 4.4, p:300. In: Von Schuckmann K, Le Traon PY, Alvarez-Fanjul E, Axell L, Balmaseda M, Breivik LA, Brewin RJW, Bricaud C, Drevillon M, Drillet Y, Dubois C , Embury O, Etienne H, García-Sotillo M, Garric G, Gasparin F, Gutknecht E, Guinehut S, Hernandez F, Juza M, Karlson B, Korres G, Legeais JF, Levier B, Lien VS, Morrow R, Notarstefano G, Parent L, Pascual A, Pérez-Gómez B, Perruche C, Pinardi N, Pisano A, Poulain PM , Pujol IM, Raj RP, Raudsepp U, Roquet H, Samuelsen A, Sathyendranath S, She J, Simoncelli S, Solidoro C, Tinker J, Tintoré J, Viktorsson L, Ablain M, Almroth-Rosell E, Bonaduce A, Clementi E, Cossarini G, Dagneaux Q, Desportes C, Dye S, Fratianni C, Good S, Greiner E, Gourrion J, Hamon M, Holt J, Hyder P, Kennedy J, Manzano-Muñoz F, Melet A, Meyssignac B, Mulet S, Nardelli BB, O'Dea E, Olason E, Paulmier A, Pérez-González I, Reid R, Racault MF, Raitsos DE, Ramos A, Sykes P, Szekely T, Verbrugge N. 2016. The Copernicus Marine Environment Monitoring Service Ocean State Report, Journal of Operational Oceanography. 9 (sup2): 235-320. http://dx.doi.org/10.1080/1755876X.2016.1273446

Pérez Gómez B., De Alfonso M., Zacharioudaki A., Pérez González I., Álvarez Fanjul E., Müller M., Marcos M., Manzano F., Korres G., Ravdas M., Tamm S. 2018. Sea level, SST and waves: extremes variability. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, Chap. 3.1, s79–s88, DOI: https://doi.org/10.1080/1755876X.2018.1489208

Woodworth, P. L., and D. L. Blackman, 2004: Evidence for Systematic Changes in Extreme High Waters since the Mid-1970s. *J. Climate*, 17, 1190–1197, https://doi.org/10.1175/1520-0442(2004)017<1190:EFSCIE>2.0.CO;2.